

# **Requirement Analysis Document (RAD)** - Task 1

Project Title: **AJIO Products Order Booking Database System**

Student Name: **Harish SP**

Register Number: ASML25005

---

## **1. Executive Project Summary**

Managing product orders, inventory, and customer data in real-time requires a robust transaction system. When a customer places an order on AJIO, the system must instantly verify product availability, reserve the items from inventory, process payment securely, and generate order confirmations with tracking details.

To handle this efficiently, AJIO's order booking system relies on a Relational Database Management System (**RDBMS**) like MySQL or PostgreSQL. A relational database is essential because it ensures data integrity and supports ACID (Atomicity, Consistency, Isolation, Durability) properties. For example:

- \* **\*\*Atomicity (A):\*\*** When a customer places an order, the system must both deduct the product quantity from inventory and record the order details. If the inventory update fails, the entire transaction rolls back, preventing orders that cannot be fulfilled.
- \* **\*\*Consistency (C):\*\*** The system maintains valid data states. For instance, product prices cannot be negative ( $\text{Price} \geq 0$ ), and an order cannot be created for a product that does not exist in the inventory.

---

## **2. Core Business Domains & Features**

Based on the analysis of AJIO's e-commerce operations, the database requirements are organized into four functional modules:

### **2.1 Customer & Account Management**

This module handles customer profiles, login credentials, contact details, and account preferences.

#### **Business Rules:**

- \* Every customer must register with a unique email address and mobile number.
- \* Customers are assigned a unique Customer ID upon registration.
- \* A customer can save multiple delivery addresses (Home, Office, etc.).
- \* Customers can exist in the system without having placed any orders yet.

- \* Account status can be Active or Inactive.

## 2.2 Product & Inventory Management

This module manages product catalogs, categories, and real-time inventory tracking.

### Business Rules:

- \* Each product belongs to a specific Category (e.g., "Men's Clothing", "Women's Footwear", "Kids' Accessories").
- \* Products have attributes like Name, Description, Brand, Color, Size, Price, and Quantity Available.
- \* Inventory quantities are updated automatically with each successful order.
- \* Products can be marked as "Out of Stock" when inventory reaches zero.
- \* A product can have multiple images associated with it.

## 2.3 Order & Cart Management

This module captures customer orders, shopping cart items, and the order lifecycle.

### \*\*Business Rules:

- \* Each order is identified by a unique Order ID.
- \* An order is linked to exactly one customer.
- \* A single order can contain multiple products (one-to-many relationship).
- \* The price recorded in order items must reflect the price at the time of purchase, even if the product price changes later.
- \* Orders follow a clear lifecycle:

***Cart → Order Placed → Payment Pending → Payment Confirmed → Shipped → Delivered → Cancelled***

## 2.4 Payment & Shipping Operations

This module manages payment processing and shipment tracking.

### Business Rules:

- \* Each order has one payment record (1:1 relationship).
- \* Order status cannot progress to "Confirmed" until payment is marked as Successful.
- \* Shipping address must be validated before shipment.
- \* Once shipped, a tracking ID is generated and linked to the order.
- \* Order status can only be "Delivered" after successful delivery confirmation.

---

## 3. Actor Profiles & System Interactions

To ensure security and appropriate access, different users have distinct database privileges. Four primary actors are defined:

1. Customer (Shopper)

- \* **Read Access:** Product catalogs, prices, order history, delivery status.
- \* **Write/Edit:** Profile information, delivery addresses, cart items, order cancellations, product reviews.
- \* **Safety Boundaries:** Cannot view other customers' details, modify product prices, or change inventory.

## 2. Inventory Manager

- \* **Read Access:** Stock levels, product performance, order volume.
- \* **Write/Edit:** Add new products, update inventory quantities, modify prices, mark products as discontinued.
- \* **Safety Boundaries:** Cannot view customer payment details, cancel confirmed orders, or modify shipping information.

## 3. Logistics Coordinator

- \* **Read Access:** Customer shipping details, order items, order status.
- \* **Write/Edit:** Update order status to "Shipped", "In Transit", "Delivered", generate tracking IDs.
- \* **Safety Boundaries:** Cannot view financial records, alter product details, or change order prices.

## 4. System Administrator

- \* **Read/Write:** Complete CRUD (Create, Read, Update, Delete) access across all tables.
- \* **Safety Measures:** All admin actions are logged for audit and fraud prevention.

---

## **4. Analytical Reporting Requirements**

The database must support analytical queries for business optimization and reporting.

### Report 1: Low Stock Alerts

- \* **Business Value:** Identifies products approaching zero inventory to trigger reordering.
- \* **Logical Query Goal:** Find all products where stock quantity is less than the reorder threshold (e.g., < 10 units). List Product Name, Category, Current Stock, and Reorder Status.

### Report 2: Top-Selling Products Report

- \* **Business Value:** Helps identify popular products for marketing and stock planning.
- \* **Logical Query Goal:** Retrieve the top 10 products ranked by total quantity sold in the last 3 months. Show Product Name, Category, Total Sales Volume, and Revenue Generated.

### Report 3: Customer Purchase Behavior Analysis

- \* **Business Value:** Identifies high-value customers for loyalty programs and targeted offers.
- \* **Logical Query Goal:** For the last 6 months, list customers with total spending above ₹10,000. Include Customer Name, Email, Total Orders, and Total Amount Spent.

---

## 5. Non-Functional Requirements & Database Constraints

### Data Constraints

- \* Product prices must be positive values ( $\text{Price} > 0$ ).
- \* Stock quantity cannot be negative ( $\text{Stock} \geq 0$ ).
- \* Customer ratings must be integers between 1 and 5.
- \* Order quantities must be positive integers.

### Concurrency Controls

- \* When two customers add the same product to their cart simultaneously, the system must use row-level locks.
- \* The first customer's transaction reserves the inventory; the second sees the updated availability.
- \* Prevents overselling of products.

### Historical Auditing & Safe Deletion

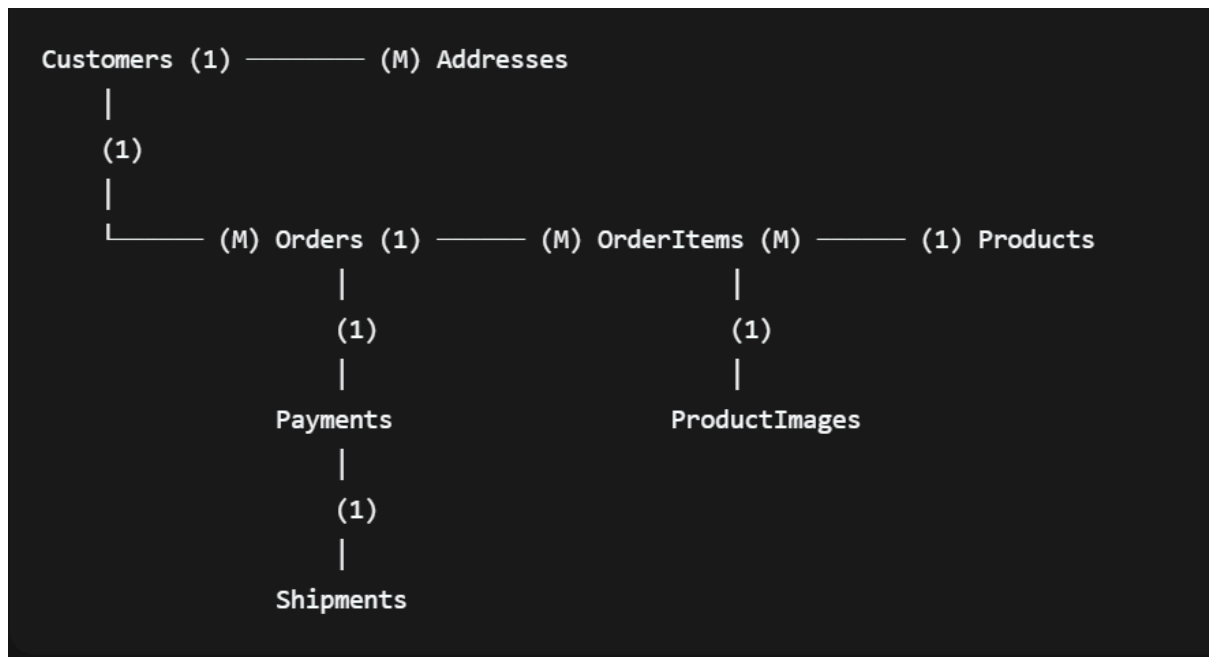
- \* Customer accounts are not permanently deleted; they are marked as "Inactive" and personal information is anonymized for privacy compliance.
- \* Past order records are retained for financial auditing and reporting.
- \* All price changes and inventory updates are logged for historical tracking.

---

## 6. Database Tables Overview

| Table Name    | Key Fields                                          | Purpose                       |
|---------------|-----------------------------------------------------|-------------------------------|
| Customers     | CustomerID, Name, Email, Password, Phone            | Store customer information    |
| Addresses     | AddressID, CustomerID, Street, City, State, Pincode | Store delivery addresses      |
| Categories    | CategoryID, CategoryName, Description               | Product categories            |
| Products      | ProductID, Name, Brand, CategoryID, Price, Stock    | Product catalog and inventory |
| ProductImages | ImageID, ProductID, ImageURL                        | Product images                |
| Cart          | CartID, CustomerID, ProductID, Quantity             | Shopping cart items           |
| Orders        | OrderID, CustomerID, OrderDate, Status, TotalAmount | Order header                  |
| OrderItems    | OrderItemID, OrderID, ProductID, Quantity, Price    | Order line items              |
| Payments      | PaymentID, OrderID, Amount, PaymentDate, Status     | Payment transactions          |
| Shipments     | ShipmentID, OrderID, TrackingID, DeliveryDate       | Shipping details              |
| Reviews       | ReviewID, CustomerID, ProductID, Rating, Comment    | Product reviews               |

## 7. Entity Relationship Overview



---

## 8. Key Business Rules Summary

- \* One customer can have multiple delivery addresses.
- \* One order belongs to exactly one customer.
- \* One order can contain multiple products through `OrderItems`.
- \* One product can appear in multiple orders.
- \* Each order has exactly one payment record.
- \* Each order has exactly one shipment record.
- \* One customer can write multiple product reviews.
- \* One product can receive multiple reviews.

-----X-----X-----X-----